



Per cgroup Congestion Detection and Fairness Enforcement

V S Susi Krishna | CS23BTECH11062

M Sai Manjunath | CS23BTECH11035

Sathvik Bhavanam | CS23BTECH11056

Adityanath M | CS23BTECH11032

Vikranth Ram | ES23BTECH11024

Sathvik J | ES23BTECH11020

Background

Background



Linux Control Groups:

- Modern Linux systems run multiple applications simultaneously.
- These applications belong to different cgroups, which are used to isolate and control resource usage such as CPU, memory, and Network bandwidth.

Local Congestion:

- When many processes generate network traffic, they share a common outgoing interface.
- Under heavy load, this interface becomes congested, causing latency spikes, packet drops, and unfair resource usage.

FQ_CoDel - Flow Queuing Controlled Delay:

- Linux provides queue disciplines like fq_codel, which:
- Reduce bufferbloat
- Maintain per-flow fairness
- Handle latency through AQM (Active Queue Management).

Problem Statement

Problem Statement



Ensure fair bandwidth usage among cgroups during congestion

- In Multi-tenant system
 - one cgroup may generate disproportionate traffic during congestion
- FQ_CoDel ensures fairness among flows not cgroups

Sub-problems:

- Detect congestion on the target interface.
- Temporarily enforce fairness across cgroups whenever congestion happens.
- Avoid modifying the Linux kernel - use eBPF to extend behavior safely.

System Design and Architecture



Detecting

Detecting local congestion

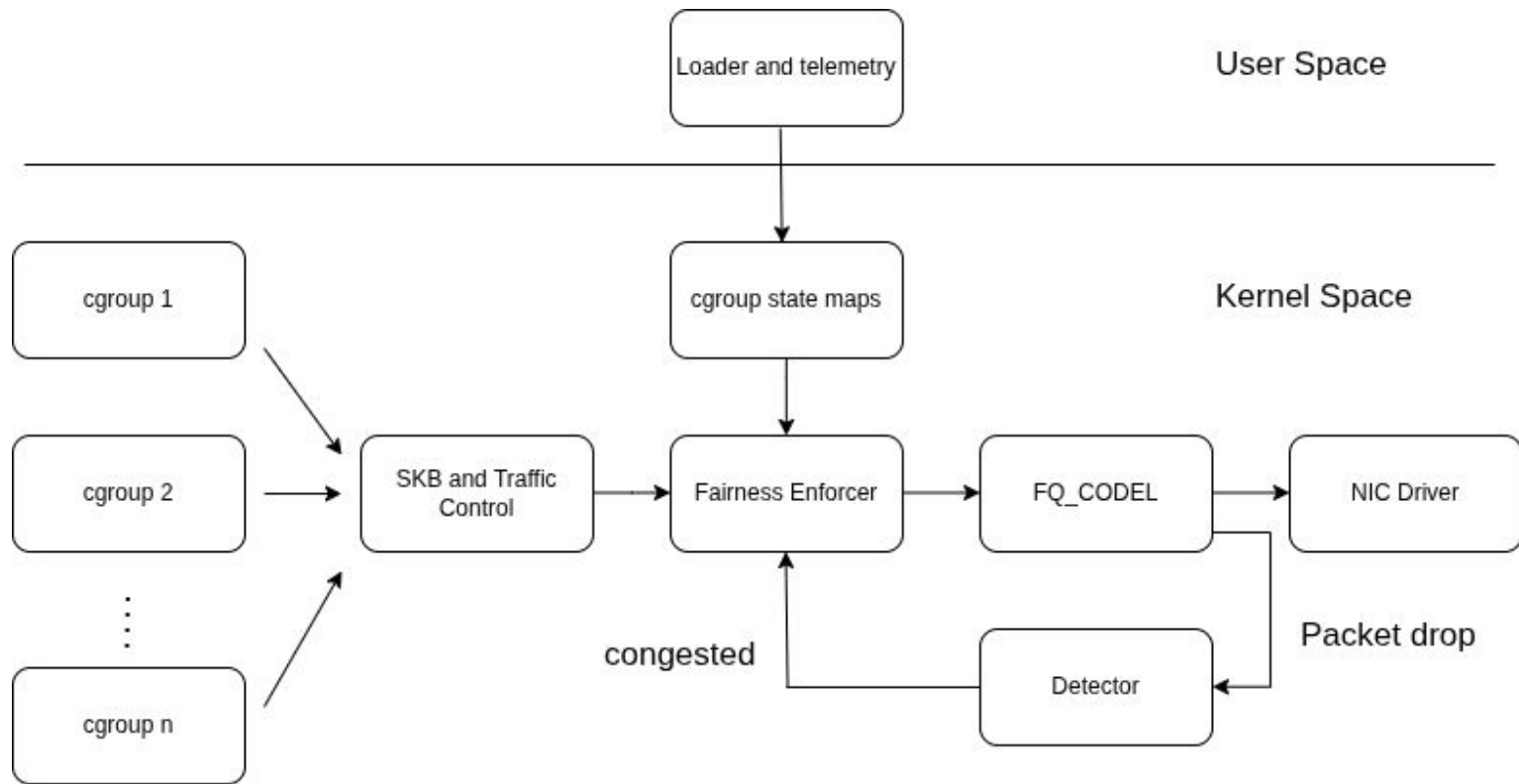
Use existing FQ_CoDel infrastructure to detect congestion.

Enforcing

Enforcing fairness among cgroups.

After congestion has been detected limit further traffic from entering FQ_CoDel

Packet Flow Diagram



Implementation

Congestion Detector (detect.c)



Detect when congestion is occurring on the chosen interface.

How it works:

- Hooks into tracepoint - kfree_skb
- Reads skb->dev->ifindex
- If drop occurred on our interface → marks congestion in a shared BPF map
- Sets a fairness window (e.g., 2 seconds):

```
congestion_map.active = 1
```

```
congestion_map.expiry_ns = now + 2 seconds
```

This makes detection fast, kernel-independent, and reliable across kernels.

Fairness Enforcer (enforce . c)



Enforce temporary bandwidth fairness among cgroups during congestion.

How it works:

- Runs at TC egress
- For each packet:
 - Reads current cgroup ID
 - Checks if congestion is active
 - Uses per-cgroup token buckets to limit rate
 - Allows or drops the packet accordingly
- Updates per-cgroup counters:
 - packets passed
 - bytes passed
 - packets dropped
 - bytes dropped

This ensures that during congestion, aggressive cgroups cannot overwhelm others.

User-space Loader + Tools



Loader (C program):

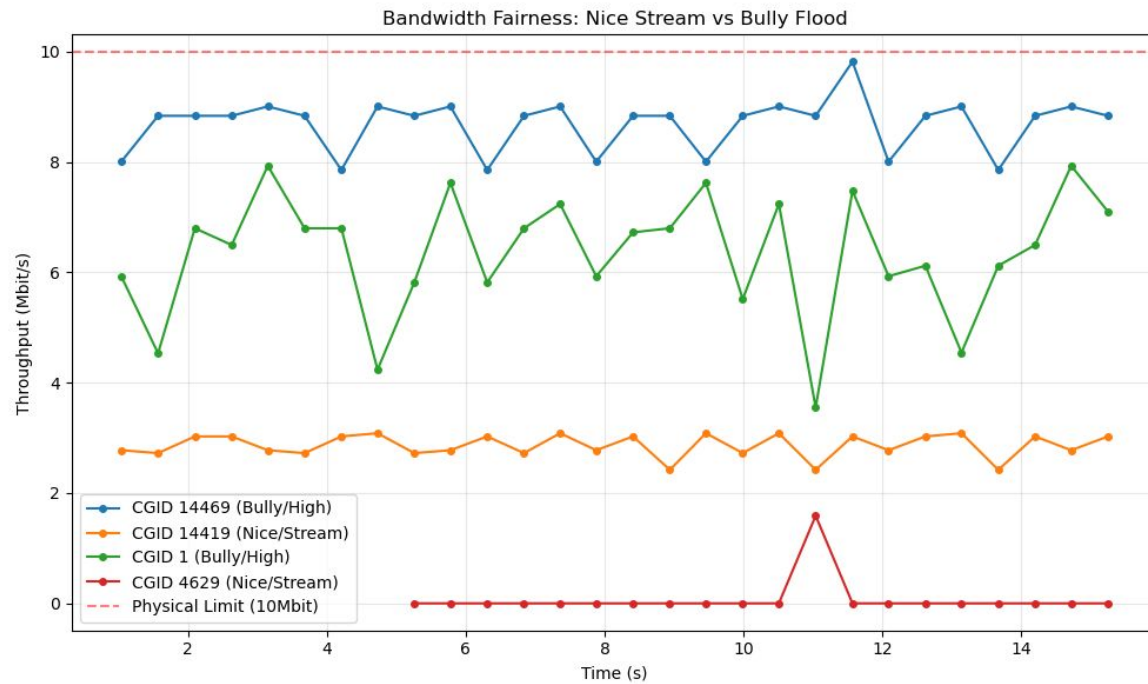
- Loads both eBPF programs
- Attaches tracepoint and TC hook
- Writes target interface ifindex into configuration map

CLI Tools:

- cli — continuously manages BPF state
- run_test.sh — runs automated experiments
- plot_metrics.py — graphs throughput, drops, congestion state

Evaluation

Evaluation



Future Work



- Extended the fairness to within a cgroup - between TCP and UDP



Thank You!